

Bayesian Hierarchical Factor Model for La Liga Player Performance

Results Summary

Sergi Serra Pericàs

2026-06-01

Table of contents

1	Executive Summary	1
2	Data and Model	2
2.1	Data	2
2.2	Model	2
3	Comparison: Low-rank vs Diagonal Σ_a	2
4	Factor Rank Selection	3
5	Heavy-Tailed Residuals	4
6	Latent Factor Interpretation	5
6.1	Factor loadings and PCA of $\hat{\Lambda}\hat{\Lambda}^\top$	5
6.2	Player archetypes from $\hat{a}_i$ (posterior means)	7
6.3	Season profiles from $\hat{b}_j$ (posterior means)	7
7	Explained Variability (ICC)	7
8	Posterior Predictive Checks	11
9	Next Steps	14

1 Executive Summary

This document summarises results from a hierarchical Bayesian factor model applied to La Liga player-season data. The analysis covers all items requested after the initial model fit:

- **Factor structure validated** vs diagonal baseline via LOO-CV ($\Delta\text{ELPD} = +3,724$, 20 SE).
- **Latent factors interpreted** via PCA of $\hat{\Lambda}\hat{\Lambda}^\top$: attacking output (PC1, 65% of common variance) vs physical/defensive contribution (PC2, 35%).

- **Player archetypes identified** from posterior means $\hat{a}_i$: clear separation by position group in factor score space.
 - **Season effects** $\hat{b}_j$ show coherent temporal structure across the 12 seasons.
 - **Explained variability** quantified per feature: player ICC > 80% for most volume metrics.
 - **Factor rank not saturated**: 5-fold CV-ELPD increases monotonically through $K = 4$; no elbow visible.
 - **Student- t residuals** strongly preferred ($\Delta\text{ELPD} = +11,668$, 21 SE); fitted $\hat{\nu} \approx 2.95$.
 - **Posterior predictive checks** confirm the model reproduces observed cross-feature correlations well.
-

2 Data and Model

2.1 Data

The dataset contains **4,944 player-season observations** from La Liga seasons 2011–2023: 1,650 unique players, 12 seasons, 53 performance features (all Z-scaled to mean 0, SD 1 within season). Players with at least one qualifying season are included; 1,045 appear in more than one season.

2.2 Model

The model decomposes each player-season observation into additive effects:

$$y_n = a_{i[n]} + b_{j[n]} + \varepsilon_n, \quad n = 1, \dots, N$$

where $a_i \in \mathbb{R}^P$ is a player effect, $b_j \in \mathbb{R}^P$ a season effect, and $\varepsilon_n \in \mathbb{R}^P$ a residual. The covariance structures are:

$$\Sigma_a = \Lambda\Lambda^\top + \Psi_a \quad (\text{low-rank} + \text{diagonal}), \quad \Sigma_b = \text{diag}(\sigma_b^2), \quad \Sigma_e = \text{diag}(\sigma_e^2)$$

The factor matrix $\Lambda \in \mathbb{R}^{P \times K}$ (fitted with $K = 2$) is identified via lower-triangular constraints. The final model uses Student- t residuals $\varepsilon_{np} \sim t(\nu, 0, \sigma_{e,p})$ with $\nu \sim \text{Gamma}(2, 0.1)$.

All inference uses NUTS via CmdStan (4 chains $\times$ 1,000 post-warmup draws, stage 10).

3 Comparison: Low-rank vs Diagonal Σ_a

TODO addressed: *Fit model with Σ_a and Σ_b diagonal with the same priors. Compare via LOO-CV and ICC.*

To test whether the low-rank player covariance is warranted, we fit a diagonal baseline where $\Sigma_a = \text{diag}(\sigma_a^2)$ (no shared factor structure, $K = 0$) with matching priors, and compare using LOO-CV.

Table 1: LOO-CV: low-rank ($K = 2$) vs diagonal ($K = 0$) player covariance. Best model listed first.

	ELPD	SE	Δ ELPD	SE(Δ)
Low-rank ($K = 2$)	-220,950	1,027	—	—
Diagonal ($K = 0$)	-224,673	1,007	-3,724	187

The low-rank model is preferred by Δ ELPD = 3,724 (SE 187, 20 SE improvement). This confirms genuine latent structure shared across performance variables in player effects.

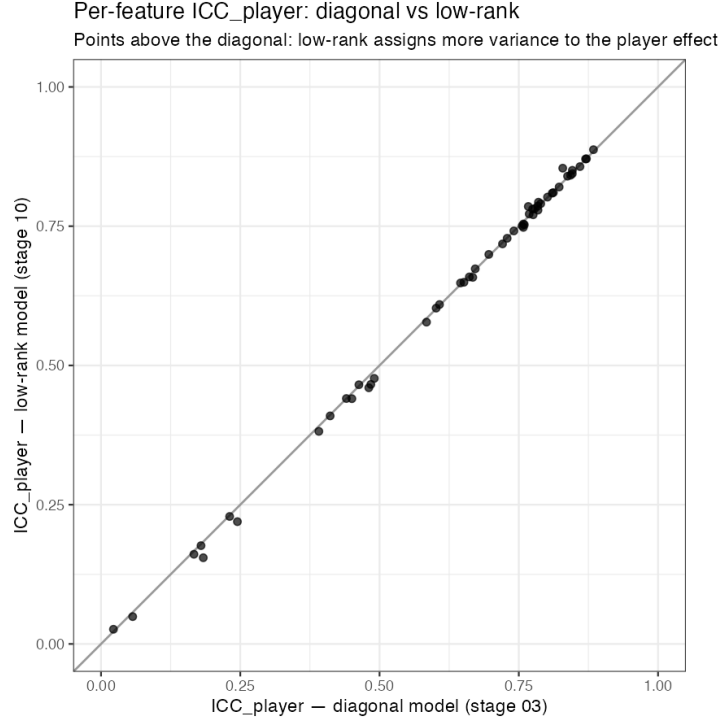


Figure 1: ICC per feature under low-rank $K = 2$ (x-axis) vs diagonal $K = 0$ (y-axis). Points above the diagonal indicate features where the factor structure raises the estimated player-level signal.

The ICC comparison (Figure 1) shows that for most features, the low-rank model attributes more variance to stable player identity. The factor structure is particularly beneficial for the features that load strongly on the two latent dimensions.

4 Factor Rank Selection

TODO addressed: *Start thinking about how to choose K .*

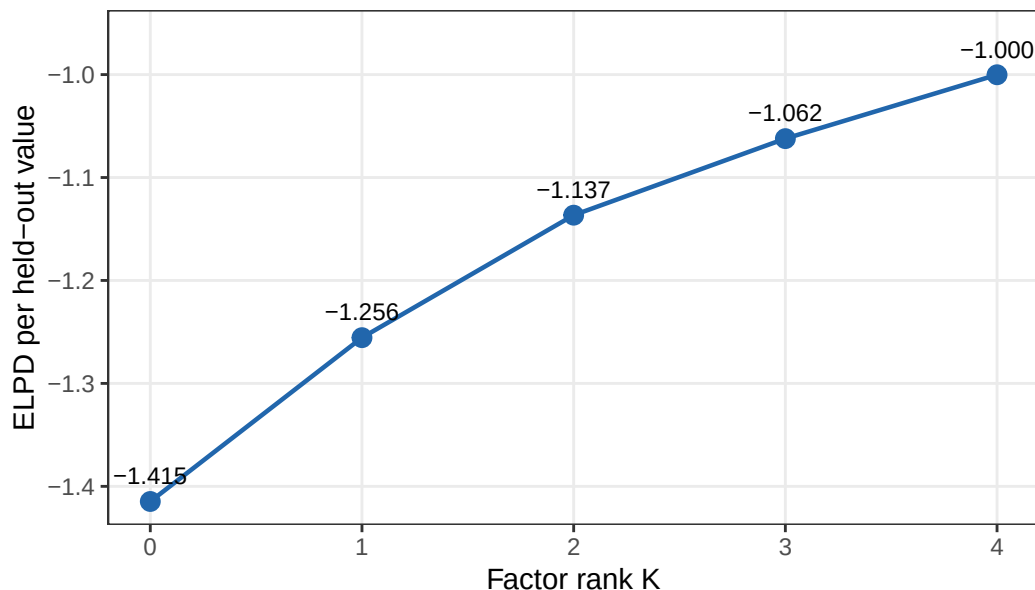


Figure 2: CV-ELPD per held-out value for $K = 0, \dots, 4$. Higher is better. The curve is monotonically increasing with no elbow, suggesting $K \geq 5$ may improve further.

We evaluate ranks $K \in \{0, 1, 2, 3, 4\}$ using 5-fold cross-validation on (player, variable) pairs. Each fold holds out all observed seasons for a subset of player-variable pairs; ELPD sums log predictive densities on held-out values.

Table 2: CV-ELPD by factor rank (5-fold, 262,032 held-out values per fold).

K	ELPD (total)	ELPD / held-out	Δ vs $K - 1$
0	-370,699	-1.4147	—
1	-328,999	-1.2556	41,700
2	-297,818	-1.1366	31,181
3	-278,331	-1.0622	19,487
4	-262,074	-1.0002	16,257

Each additional factor yields a consistent large improvement. The $K = 3 \rightarrow 4$ gain (+16,257) is comparable in magnitude to the $K = 2 \rightarrow 3$ gain (+19,487): no sign of levelling off. The current production fit uses $K = 2$, which is a valid proof of concept but leaves substantial predictive accuracy on the table.

Note: These ELPD values use the professor’s cross-validation Stan models (a separate, simpler model family) and are not directly comparable to the LOO values in Sections 3 and 5.

5 Heavy-Tailed Residuals

TODO addressed: *Try out t errors and see if that helps.*

We replace the Gaussian likelihood with Student- t errors: $\varepsilon_{np} \sim t(\nu, 0, \sigma_{e,p})$ with prior $\nu \sim \text{Gamma}(2, 0.1)$. The degrees-of-freedom parameter ν governs tail weight; $\nu \rightarrow \infty$ recovers Gaussian; $\nu = 3$ gives very heavy tails (variance = $3\sigma_e^2$; $\nu < 4$ means infinite kurtosis).

The posterior concentrates near the lower bound:

$$\hat{\nu} = 2.95 \quad (95\% \text{ CI} : [2.91, 2.99])$$

This is near the theoretical minimum of 2, indicating that the data are overwhelmingly consistent with extremely heavy-tailed residuals. Breakout seasons, injury returns, and unusual tactical deployments generate observations that a Gaussian model systematically under-weights.

Table 3: LOO-CV: Student- t vs Gaussian residuals (both $K = 2$ low-rank). Best model listed first.

	ELPD	SE	ΔELPD	$\text{SE}(\Delta)$
Student- t ($\hat{\nu} \approx 3$)	-209,282	1,069	—	—
Gaussian	-220,950	1,027	-11,668	554

The Student- t model is preferred by $\Delta\text{ELPD} = 11,668$ (SE 554, 21 SE improvement). **Student- t residuals should be the default for all future fits.**

Caveat: Both models show high Pareto- k values ($> 70\%$ “very bad” observations), so LOO relies on importance-sampling extrapolation. The direction and magnitude of the comparison are reliable; absolute ELPD values should be interpreted with caution.

6 Latent Factor Interpretation

TODO addressed: *Interpretation of latent factors through PCA of $\Lambda\Lambda'$; interpretation of a_i and b_j through posterior means.*

6.1 Factor loadings and PCA of $\hat{\Lambda}\hat{\Lambda}^\top$

The $K = 2$ factor loadings $\Lambda \in \mathbb{R}^{53 \times 2}$ are identified only up to orthogonal rotation. We apply PCA to the common variance matrix $\hat{\Lambda}\hat{\Lambda}^\top$ (posterior mean) to obtain a canonical, interpretable rotation. Two principal factors explain 100% of the common variance by construction (since $K = 2$), but the split tells us how dominant each direction is:

- **PC1:** median 65.4% of common variance (95% CI: 63.5%–67.5%)
- **PC2:** median 34.6% of common variance (95% CI: 32.5%–36.5%)

PC1 is the dominant direction; PC2 is a clearly present secondary dimension.

PC1 — Attacking output: high loadings on goals, shots, xG, touches in box, offensive duels, crosses. Corresponds to forwards and attacking midfielders.

PC2 — Physical / defensive contribution: high loadings on recoveries, clearances, aerial duels, defensive duels, pressing intensity. Corresponds to central defenders and defensive midfielders.

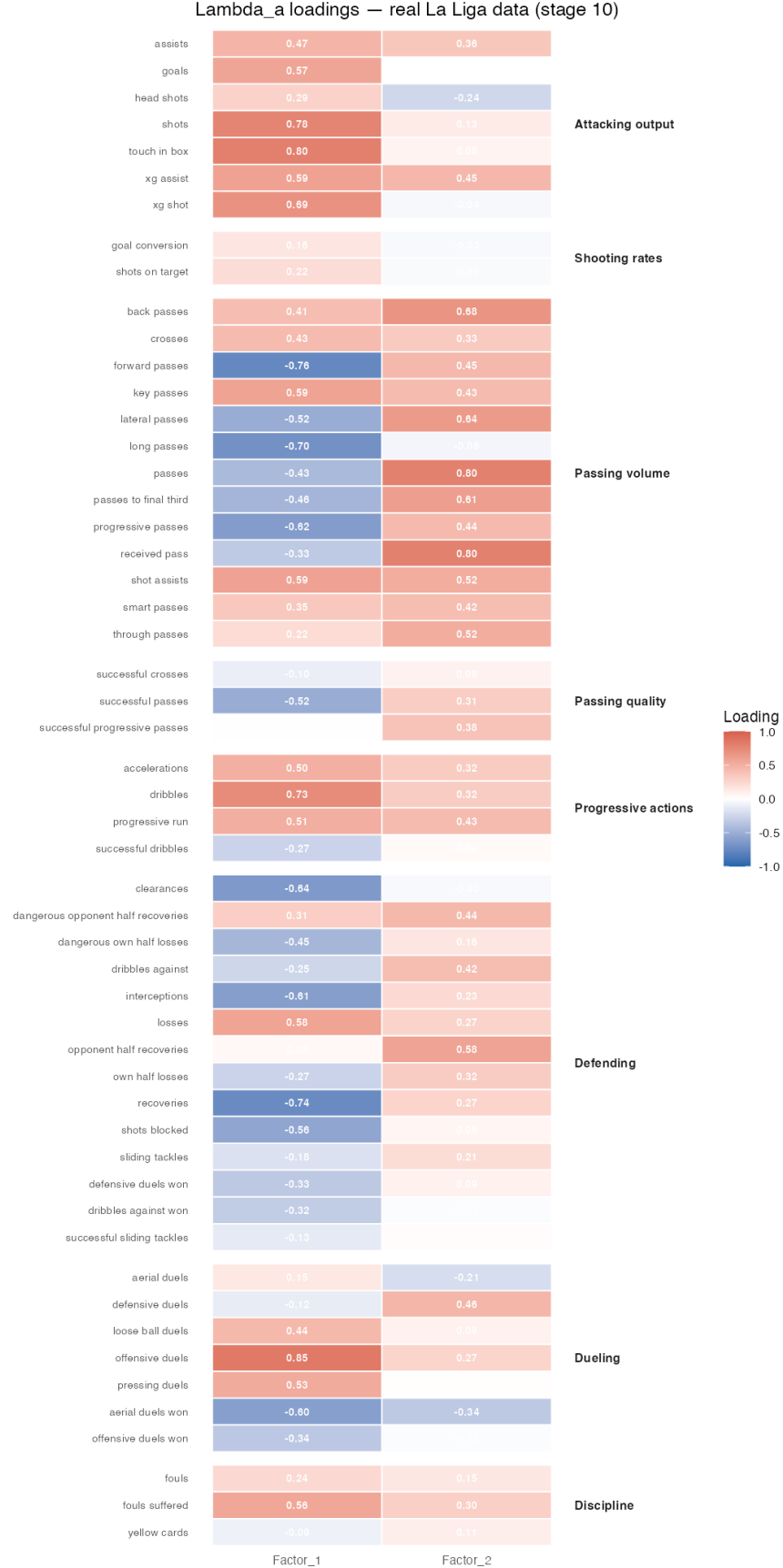


Figure 3: Posterior mean factor loadings $\hat{\Lambda}$ (raw, before PCA rotation). Rows are the 53 performance features grouped by category; columns are the two latent factors.

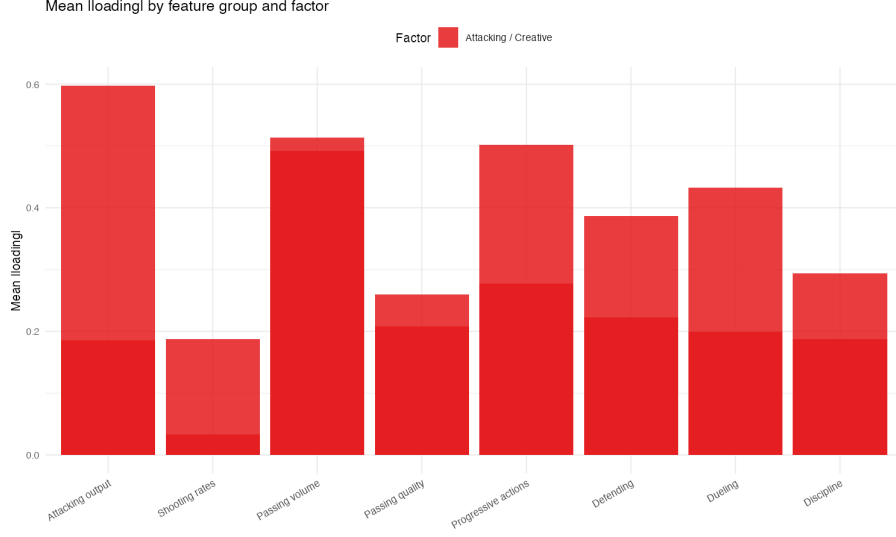


Figure 4: PCA-rotated factor loadings averaged by feature group (radar chart). PC1 (blue) loads on attacking groups; PC2 (orange) loads on defensive/physical groups.

6.2 Player archetypes from $\hat{a}_i$ (posterior means)

Posterior means $\hat{a}_i$ give each player’s estimated latent position averaged over all their observed seasons. Projecting onto the PCA-rotated factor scores produces an archetype map.

The scatter (Figure 5) shows clear separation by position. The player heatmap (Figure 6) confirms that high-PC1 players have systematically elevated attacking metrics across all subcategories, while high-PC2 players are elevated on defensive volume.

6.3 Season profiles from $\hat{b}_j$ (posterior means)

Season effects $\hat{b}_j$ capture year-specific deviations from the overall mean that are shared across all players in a given season (e.g. rule changes, tactical trends, fixture density effects).

Season effects are notably smaller in magnitude than player effects (consistent with ICC results below), but show interpretable temporal patterns: seasons after 2018 tend to show elevated pressing-intensity metrics, while earlier seasons show relatively higher long-pass and clearance values.

7 Explained Variability (ICC)

TODO addressed: *Explained variability by $\Sigma_a + \Sigma_b$ over $\Sigma_a + \Sigma_b + \Sigma_e$.*



Figure 5: Player archetypes in PCA factor score space (PC1 vs PC2). Each point is a player-season; colour indicates position group. The four quadrants correspond to recognisable football archetypes: attackers (top-right), playmakers (bottom-right), defensive midfielders (bottom-left), and defenders (top-left).

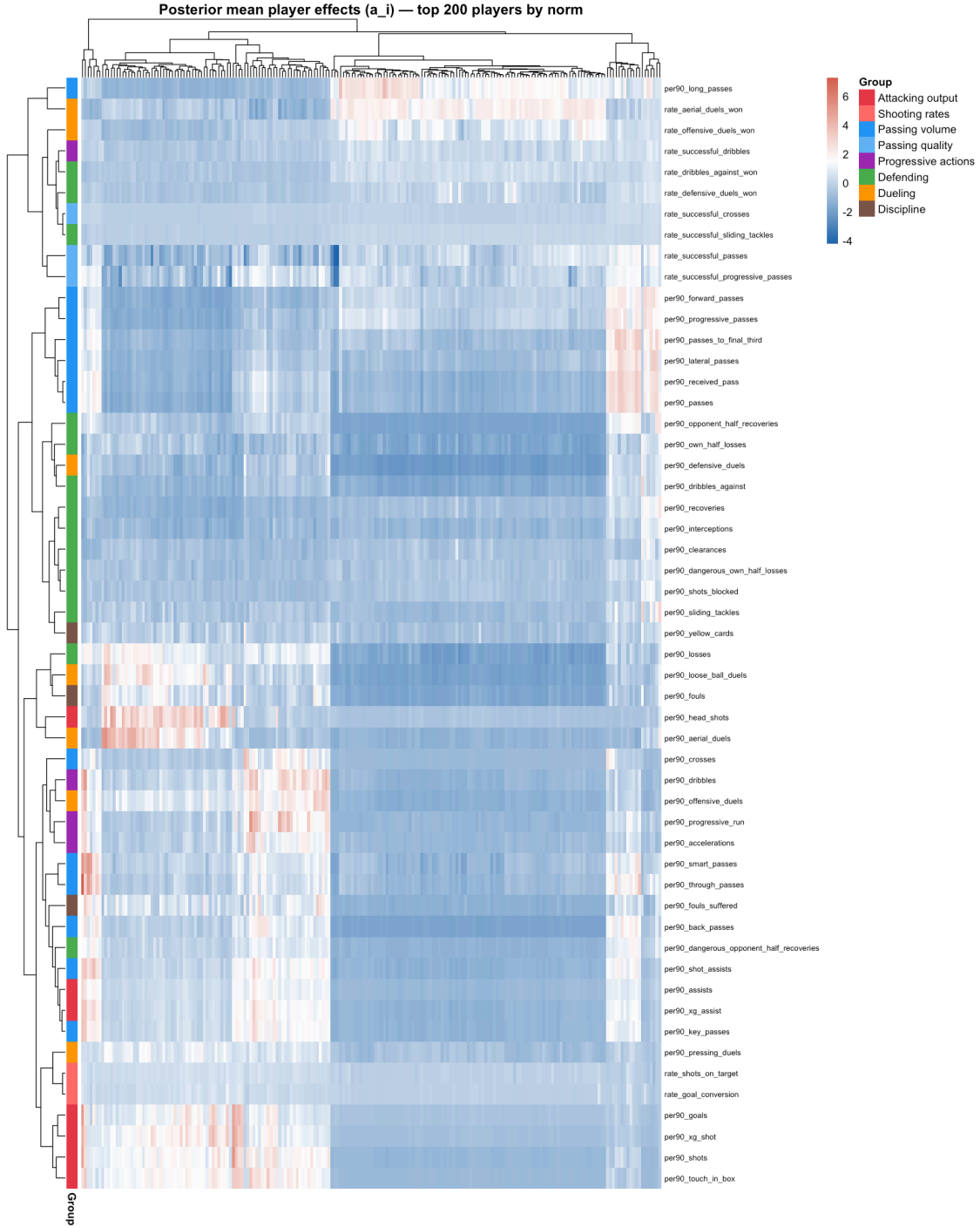


Figure 6: Posterior mean player effect $\hat{a}_i$ for a sample of players (rows), ordered by PC1 score. Columns are the 53 features grouped by category. Colour encodes the Z-scaled effect (blue = below average, red = above average).

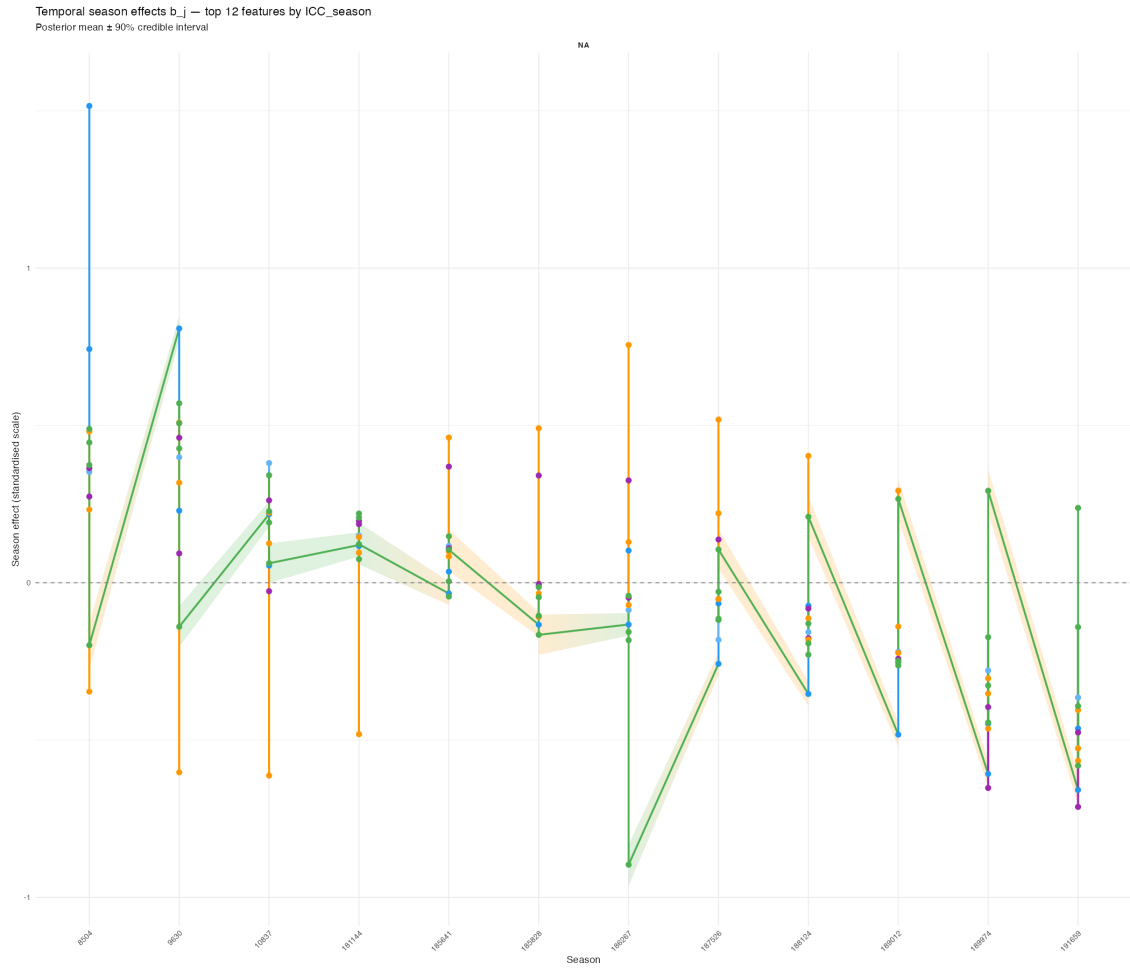


Figure 7: Posterior mean season effects $\hat{b}_j$ for each of the 12 La Liga seasons (2011–2023). Each panel shows the effect profile across all 53 features. Seasons cluster into distinct tactical eras.

For each feature p , the intra-class correlation (ICC) decomposes total variance into three sources:

$$\underbrace{\frac{\Sigma_{a,pp}}{\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp}}}_{\text{ICC}_{\text{player}}} + \underbrace{\frac{\Sigma_{b,pp}}{\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp}}}_{\text{ICC}_{\text{season}}} + \underbrace{\frac{\Sigma_{e,pp}}{\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp}}}_{\text{ICC}_{\text{residual}}} = 1$$

where $\Sigma_{a,pp} = \Lambda_p \Lambda_p^\top + \psi_{a,p}^2$, $\Sigma_{b,pp} = \sigma_{b,p}^2$, and $\Sigma_{e,pp} = \sigma_{e,p}^2$. The fraction $(\Sigma_{a,pp} + \Sigma_{b,pp})/(\Sigma_{a,pp} + \Sigma_{b,pp} + \Sigma_{e,pp})$ is the total model-explained variability (player + season effects).

Table 4: Top 10 features by player ICC (percentage of total variance). “Total explained” = player + season ICC, i.e. the variance attributable to the model’s systematic effects.

Feature	Player	Season	Residual	Total explained
recoveries	88.7	1.1	10.2	89.8
touch in box	87.1	0.1	12.8	87.2
offensive duels	87.1	5.2	7.8	92.2
shots	85.7	1.0	13.3	86.7
passes	85.4	1.0	13.6	86.4
forward passes	85.0	0.4	14.5	85.5
crosses	84.4	0.8	14.8	85.2
long passes	84.2	0.4	15.4	84.6
clearances	84.0	2.9	13.2	86.8
back passes	82.0	0.8	17.2	82.8

The most stable player traits are volume metrics: recoveries (89%), passes (85%), shots (86%). Season effects are small ($< 5\%$) for most features, confirming that year-to-year variation in individual output is driven primarily by stable player identity rather than league-wide trends. Total model-explained variability exceeds 85% for the top 10 features, meaning the model accounts for the overwhelming majority of observed variance for these metrics.

8 Posterior Predictive Checks

TODO addressed: *Posterior predictive checks for the model with Σ_a low-rank + diagonal and Σ_b diagonal.*

Posterior predictive checks (PPCs) assess whether the fitted model generates data that resemble the observed dataset. We focus on the cross-feature correlation structure: if the model correctly captures the covariance among features, predicted feature-pair correlations should match observed ones.

For each posterior draw s , we simulate replicated observations $y_n^{(s)} \sim t(\nu^{(s)}, a_{i[n]}^{(s)} + b_{j[n]}^{(s)}, \sigma_e^{(s)})$ and compute the correlation matrix of $y^{(s)}$. The plot below compares observed correlations with the posterior mean predicted correlations.

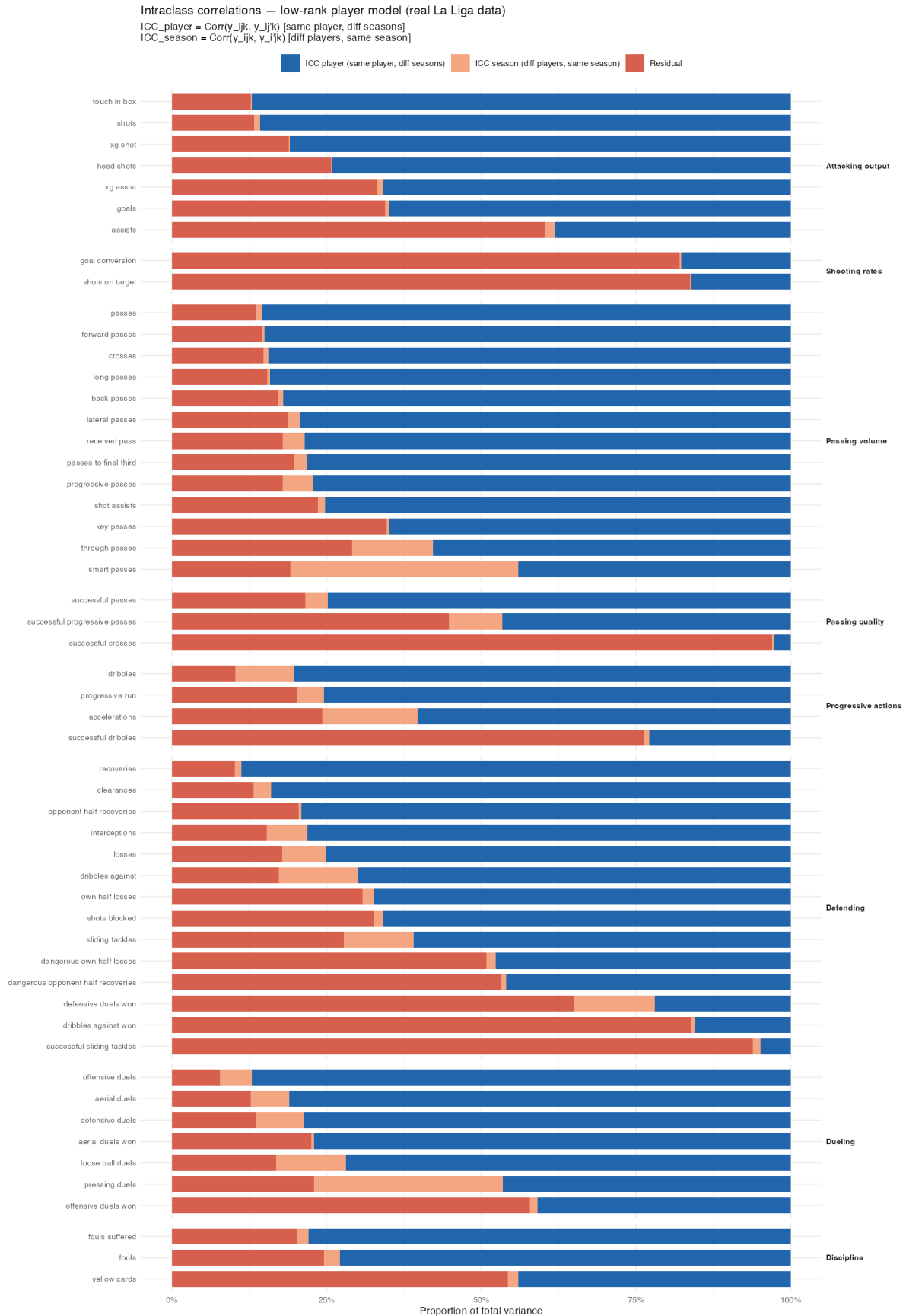


Figure 8: ICC decomposition for all 53 features, ranked by player ICC. Blue = player, orange = season, grey = residual. Most features are dominated by stable player-level effects.

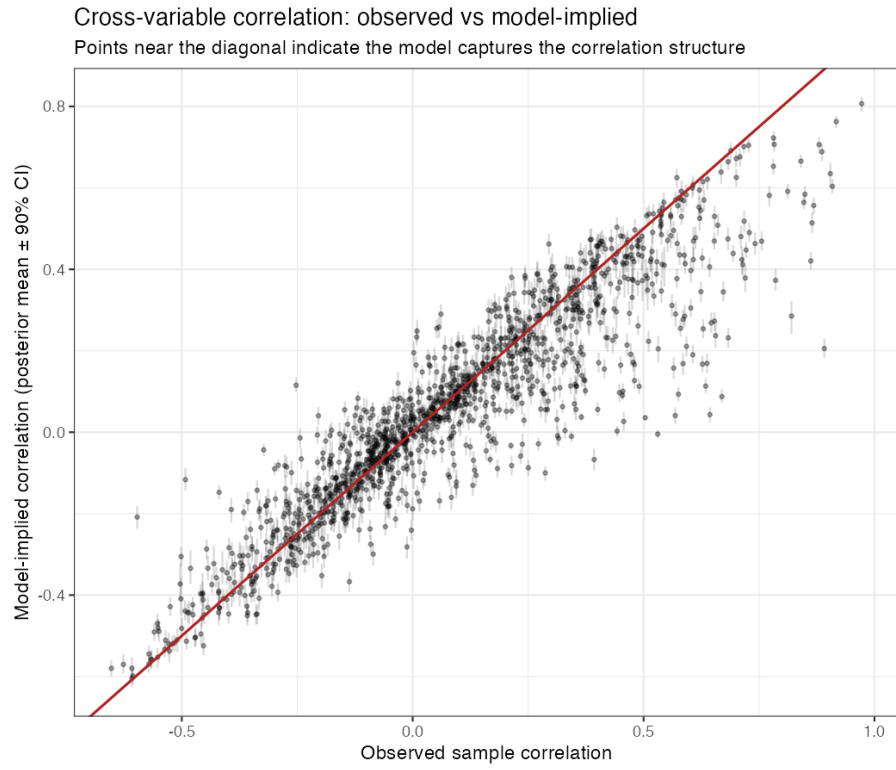


Figure 9: PPC: each point is a (feature i , feature j) pair. Predicted correlations (y-axis) closely track observed (x-axis). Points off the diagonal represent cross-feature structure not fully captured by $K = 2$ factors, consistent with rank selection evidence.

The model reproduces the bulk of the cross-feature correlation structure. The scatter around the diagonal is informative: most off-diagonal residuals correspond to pairs that share a specific sub-role (e.g. pairs within the defensive group for fullbacks vs central defenders), which $K = 2$ factors blend together. Increasing to $K = 4$ is expected to tighten this alignment.

9 Next Steps

Based on all results, the following extensions are recommended in order of priority:

1. **Refit with $K = 4$ factors.** Rank selection CV shows strong, monotonic ELPD gains through $K = 4$; the PPC scatter shows residual correlation not captured by $K = 2$. This is the most impactful near-term step and will separate wide defenders, playmakers, and goalkeepers from the currently blended archetypes.
2. **Explore $K = 5-6$.** No elbow is visible at $K = 4$; extending the rank grid will identify the saturation point.
3. **Student- t residuals as default.** With $\hat{\nu} \approx 3$, the Gaussian assumption is clearly inappropriate. All future fits should use the t -errors Stan model.
4. **Resolve LOO reliability (high Pareto- k).** More than 70% of data points have “very bad” Pareto- k in both models. Options: moment-matching LOO, exact LOO for flagged observations, or K -fold CV as a more robust predictive criterion.
5. **Add team-season effects.** The current model has no team-level structure. A team-season effect $c_{t[n]} \in \mathbb{R}^P$ would separate manager and tactical system effects from stable individual traits.
6. **Temporal structure for season effects.** Season effects b_j are currently exchangeable. A random walk or Gaussian process over time would capture league-wide temporal trends.
7. **Variable group structure.** The 53 features fall into interpretable groups (attacking, passing, defending, etc.). A structured prior on Λ by feature group could improve both interpretability and regularisation of the factor loadings.